Songwei Ge

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Education

Department of Computer Science, University of Maryland — *College Park, MD, USA*

Ph.D. in Computer Science — *Sep. 2020 - Present*

- Selected course: Foundation of Deep Learning.
- Advisor: David Jacobs.

School of Computer Science, Carnegie Mellon University — *Pittsburgh, PA, USA*

M.S. in Computational Biology — *Aug. 2018 - May. 2020*

- GPA: 4.15/4.00
- Selected courses: Machine Learning for Ph.D., Convex Optimization, Probabilistic Graphical Model.
- Advisors: Barnabás Póczos, Eunsu Kang and Ziv Bar-Joseph.

School of Information, Renmin University of China — *Beijing, China*

B.S. in Computer Science and Technology — *Sep. 2014 – Jul. 2018*

- Thesis Advisors: Zhicheng Dou and Ji-Rong Wen. (**Best thesis award** in Computer Science.)

HillHouse Academy, Renmin University of China — *Beijing, China*

B.S. in Finance — *Sep. 2014 – Jul. 2018*

- Thesis Advisor: Wei Xu.

Research Experience

Machine Learning Department, Carnegie Mellon University — *Pittsburgh, PA, USA*

Advisors: Ziv Bar-Joseph, Zachary Lipton and Barnabás Póczos — *Jan. 2019 – Present*

- Improved the state-of-the-art alignment methods for single cell RNA-seq data with conditional domain adversarial training.
- Investigated the generalization bound in Domain Generalization with label shift.
- Studied the robustness of generic CNN-based model under statistical and adversarial distribution shift and proposed a novel regularization technique on the local predictive power to improve the overall generalization capacity.

Big Data Analytics and Intelligence Lab, Renmin University of China — *Beijing, China*

Advisors: Zhicheng Dou and Ji-Rong Wen — *Jan. 2017 – Feb. 2018*

- Proposed the first deep learning model and achieved state-of-the-art search results quality with personalization on a large-scale query log.
- Modeled long-term (whole history) and short-term (present session) user profiles using hierarchical RNN that dynamically generate personalized ranking according to the current query.

Work Experience

Facebook AI Research — *Menlo Park, CA, USA*

Advisors: Vedanuj Goswami, Larry Zitnick, Devi Parikh — *June. 2020 – Aug. 2020*

- Collected two datasets of creative sketches and each contains 10k sketches along with part annotations. Proposed DoodlerGAN – a part-based Generative Adversarial Network (GAN) – to generate unseen compositions of novel part appearances. Developed a web demo for the project.

Microsoft Xiaoice Science Team — *Beijing, China*

Advisor: Ruihua Song — *Apr. 2018 – Jul. 2018*

- Explored acoustical imagery hidden in a textual story and designed contextualized frameworks to automatically add sound effects to a radio story which achieved smaller false positives than classic methods.

Selected Publications

1. **Songwei Ge**, Vedanuj Goswami, Larry Zitnick, Devi Parikh. Creative Sketch Generation. *In Workshop on Machine Learning for Creativity and Design at NeurIPS, 2020. (Oral) In submission to ICLR 2021.*
2. **Songwei Ge**, Haohan Wang, Amir Alavi, Eric P. Xing, Ziv Bar-Joseph. Supervised Adversarial Alignment of

Single-Cell RNA-seq Data. *In the 24th International Conference on Research in Computational Molecular Biology (RECOMB), 2020*

3. Haohan Wang, **Songwei Ge**, Eric P. Xing, Zachary C. Lipton. Learning Robust Global Representations by Penalizing Local Predictive Power. *In Neural Information Processing Systems (NeurIPS), 2019.*
4. **Songwei Ge**, Curtis Xuan, Ruihua Song, Chao Zou, Wei Liu and Jin Zhou. From Text to Sound: a Preliminary Study on Retrieving Sound Effects to Radio Stories. *In ACM International Conference on Research and Development in Information Retrieval (SIGIR), 2019.*
5. **Songwei Ge***, Austin Dill*, Eunsu Kang*, Chun-Liang Li*, Manzil Zaheer and Barnabás Póczos. Developing Creative AI to Generate Sculptural Objects for Interactive Art Installation. *In Inter-Society for the Electronic Arts (ISEA), 2019*
6. **Songwei Ge**, Zhicheng Dou, Zhengbao Jiang, Jian-Yun Nie, Ji-Rong Wen. Personalizing Search Results Using Hierarchical RNN with Query-aware Attention. *In ACM International Conference on Information and Knowledge Management (CIKM), 2018*

Selected Honors & Awards

2019-2020	**CMU MSCB Academic Scholarship**	*Pittsburgh, PA*
Jun 2018	**Outstanding Undergraduate Thesis Award** (Top 1% in RUC & Top 1 in the Computer Science)	*Beijing, China*
Jan 2017	**Meritorious Winners,** American Mathematical Contest in Mathematical Modeling	*Beijing, China*
2014-2018	**RUC School of Information Scholarship**	*Beijing, China*

Teaching

2020	**Teaching Assistant** Introduction to Machine Learning	*College Park, MD*
2019	**Guest Lecturer** Art and Machine Learning	*Pittsburgh, PA*
2017	**Teaching Assistant** Advanced Algorithm Design	*Beijing China*

Skills

Programming and Deep Learning Libraries	Python, C, C++, C#, R, MATLAB, Java, Javascript, HTML, Tensorflow, PyTorch
Others	Qualified basketball referee